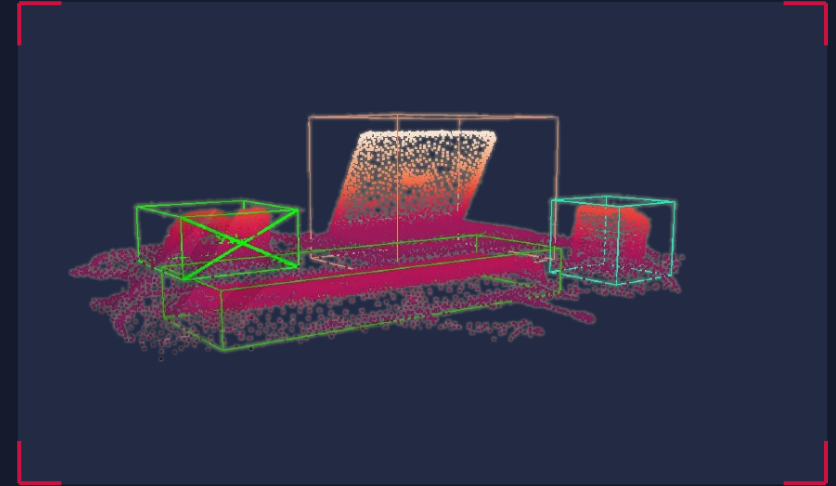


AXIOMALITY

Every robot dataset ships with a machine-checkable passport.

The verification layer for physical-AI data: a formally checked engine that scores, signs and audits every robot trajectory, so labs train on evidence instead of assumptions.

Seed round · September 2026 · Confidential



AXIOMALITY sample 429: measured point cloud with oriented 3D boxes (real data)

ISO/IEC 21838-4

Founded by a core contributor to the
TUpper top-level ontology standard

100,000

Measured 3D asset packages (photo ·
USDZ · PLY · JSON)

13 + 9

Granted patents + invention
applications

Q4 2026

Engine internal test; first paid design
partners 2027

PROBLEM

Robot teams cannot trust their data, and the flood is getting worse

Three failure modes that cost real engineering hours today, none of which produce a record anyone else can check



Before training

Demonstrations, field logs, scans and generated clips arrive with different frames, units and task labels. Coordinate errors, duplicates and mislabeled phases train silently. In a 2026 Voxel51 survey, 89% of teams blamed data for model failures and 48% cited quality problems.

“We spend 600 review hours a month and still don’t know what got through.”



After a failure

Perception, coverage, task order and physical contact all produce the same failed rollout. Engineers need the source record, a localized cause and a regression test. Today they get a video and a guess.

“Which frame, which rule, which batch? Nobody can tell us.”



Before release

New sensors, sites and synthetic mixtures invalidate earlier conclusions. There is no versioned, per-record evidence of what a model was trained on, no real-to-synthetic ratio, no consent lineage, and nothing a buyer, insurer or notified body can re-check.

“We can’t prove what we trained on, so we can’t sell it or ship it into the EU.”

\$50-200 / h

Teleoperation data, with no record of what is usable

58,000+

Datasets on the LeRobot hub (May 2026); none validated or signed

≈ 0

Direct reuse of the action layer across embodiments

0

Vendors selling a per-record, machine-checkable certificate

Sources: Voxel51 State of Visual AI survey (2026); Hugging Face (May 2026); DreamVu, Robot Training Data Companies: The 2026 Landscape; Open X-Embodiment (2023).

WHY NOW

Raw hours are going to zero, generators are flooding training, and the money has arrived

Value is moving from hours to evidence; the evidence format becomes the unit of trade

**Supply: raw data is free**

Build AI released Egocentric-1M, about one million hours of factory video under Apache-2.0 (Apr 2026). Teleoperation at \$50-200/hour has no moat. Margin survives only in semantics, verification and certification.

**Demand: synthetic data**

World Labs' Atlas (Sep 2026) generates robot-camera RGB and depth from phone video; NVIDIA's Cosmos data-factory blueprint ships a free Evaluator that scores generated clips. More generators mean more unverified data entering training, with no signed record of real-to-synthetic ratio.

**Money: budgets arrived**

Physical-AI startups raised a record \$18.6B in Q2 2026 alone. Physical Intelligence (~\$11B), Skild (\$14B), Generalist (\$3B) and Figure (\$39B) are buying data; Scale's physical-AI engine passed 150k hours. Buyers now gate their own data and need a neutral referee.

Certification	audit packs, listing evidence; what regulators and buyers accept
Verification	physics, logic, uncertainty: is the record consistent?
Semantics	what happened: objects, affordances, contacts, task phases
Raw hours	\$50-200/h (2024) → near zero (2026)

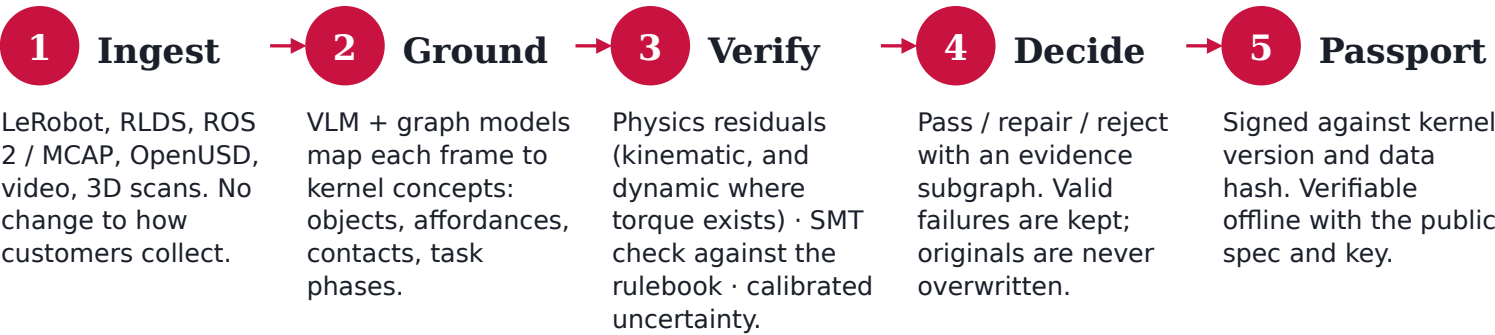
Compliance: tailwind, not thesis

EU AI Act Annex I applies to AI embedded in machinery, including ML safety components of robots, from 2 Aug 2028 (Reg. 2026/1744). We sell to the data race today; the audit pack is what the same evidence becomes in 2028.

Sources: Build AI (Apr 2026); World Labs (Sep 2026); NVIDIA (Mar 2026); PitchBook Q2 2026; company announcements; European Commission (Regulation (EU) 2026/1744).

Engine + Passport: any format in, a per-record verdict and a signed certificate out

Five stages; every episode gets pass / repair / reject with an evidence trail back to the exact frame and rule



Check	Method	Criterion	On failure
Physical consistency	Geometry, kinematics, and dynamics residual where τ / contact exist	Residual within calibrated tolerance	Repair: project to nearest feasible record, logged
Logical consistency	SMT (Z3) against kernel axioms Φ over a bounded scene graph	SAT pass / UNSAT reject / UNKNOWN surfaced	Reject: human review with diagnostic; never silently passed
Uncertainty	Conformal prediction, coverage $\geq 1 - \alpha$, reported per group	Inside prediction set	Low confidence: escalate to a more expensive check

One rulebook, compiled twice: a differentiable soft constraint for training and a decidable hard check for verification.

AXIOMALITY PASSPORT · sample

Dataset	AX-2027-0412 · kitchen tasks
Episodes	12,480 (8,140 real / 4,340 syn.)
Kernel	v1.2 · ISO/IEC 21838-4 aligned
Grounding	96.4% frames mapped (audited)
Physics	99.1% pass · 0.7% fix · 0.2% reject
Uncertainty	0.94 coverage, per group
Safety rules	ISO 13482-derived · 0 violations
Provenance	Per-episode hashes · consent
Regulatory map	EU Art. 10 / 12 · GB/Z 218.1
Signature	2027-04-12 · sha256 9f2c...a41e

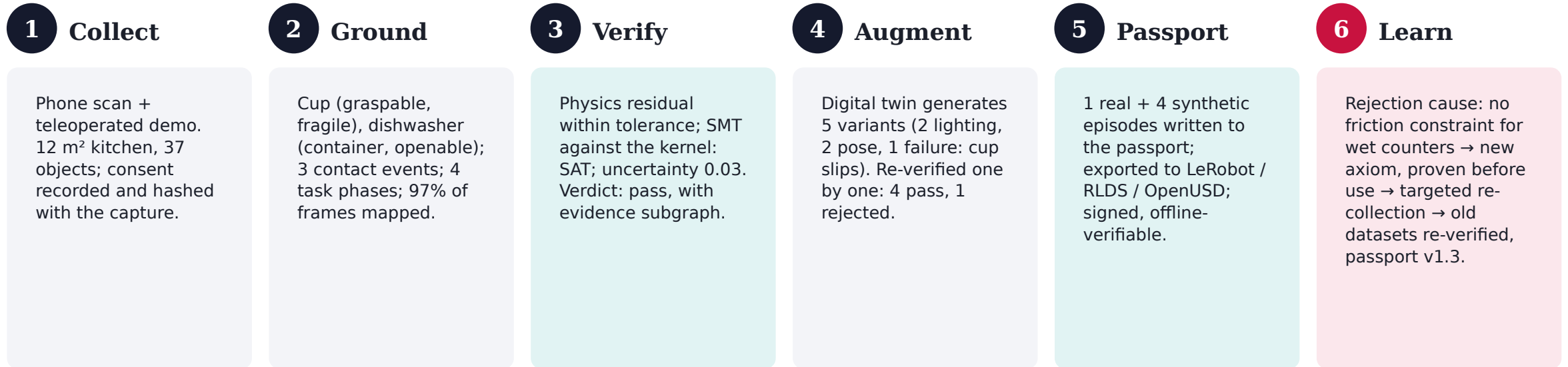
VERIFIED · offline

Product design example; values illustrate the evidence interface. Real passports on partner data under NDA from Dec 2026.

HOW IT WORKS

One episode's journey: a rejected variant becomes a new rule, and old data gets more valuable

Illustrative episode AX-2027-0412: kitchen, place a cup into a dishwasher



← Feedback: confirmed failures → new axioms → higher kernel coverage → lower verification cost per hour → collection spent only where the loop points

Once

Each new concept is axiomatized once, then reused by every episode

\$60 → \$12 / h

Fully loaded verification cost, Q4 2026 → 2030 (model target)

< \$5 / h

Marginal cost target for a re-verified synthetic episode

v1.2 → v1.3

Delivered datasets re-signed as the kernel grows

Source: AXIOMALITY product records. Cost figures are unit-economics model targets (converted at ¥7.2 per USD), not contracted.

A mathematically checked rulebook, and a flywheel that makes each hour cheaper than the last

One rulebook feeds both the passport and the runtime guardrail; the loop compounds domain knowledge competitors cannot download

Top level · TUpper (ISO/IEC 21838-4:2023)

One of the three top-level ontologies with an ISO part (with BFO and DOLCE); our founder is a core contributor. Consistency machine-checked with Prover9 / Mace4.

Physical world

Objects, parts, shape, measurement, materials, change and process (PSL lineage).

Robotics domain

Embodiment, affordances, contact and force events, kinematic chains, task phases, safety rules derived from ISO 10218 / 13482.

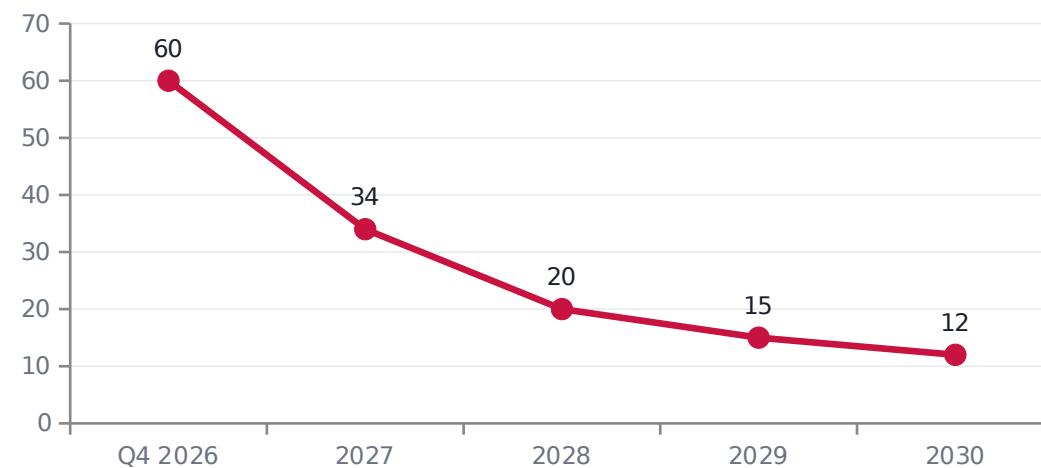
Dataset instances

Episodes, frames, sensors, lineage; exports to OpenUSD / LeRobot / RLDS.

What is rare here

Very few teams can write and machine-verify first-order axiomatizations of process, parts and physical quantities. Applying that to robot data is unusual; VLM labelling, kinematic checks and signed manifests are commodity, and we treat them that way.

Fully loaded verification cost per hour (USD, model target)



How the flywheel turns

Cost sits in the first verification of a concept, not in repeats. Customer field failures → LLM agents propose candidate rules → provers check them before release → kernel $v(n+1)$ → delivered datasets re-verified and re-signed. Human review shrinks to the reject path; what does not fall (VLM inference, prover time) is priced into the model.

Validation software for physical AI already earns premium multiples; robot learning has no such layer yet

Wedge into a countable buyer pool, meter by trajectory, then collect a per-robot royalty at fleet scale

Applied Intuition
\$15B

Simulation and validation for AVs; \$600M Series F (Jun 2025). Launched Dana for robots (Jul 2026). Proves validation tooling for physical AI is a large, high-margin category.

Foretellix
\$135M raised

Coverage-driven verification for AV; NVIDIA and Toyota Woven invested rather than built. Shows the concept, and its limit: AV-only, no per-record certificate.

Vanta
\$4.15B

SOC 2 automation; \$300M ARR (Apr 2026). The badge buyers demand becomes the product. A passport is the SOC 2 report for a robot dataset.

Stage	Construction	Annual value
Wedge (2027-28)	80-200 U.S./Canadian organizations with recurring robot data × \$150k-300k	\$12M-60M
Platform (2029-30)	500 customers × \$200k recurring, metered by verified trajectory volume	\$100M ARR
Royalty (2030+)	Runtime guardrail per robot per year; China alone forecasts 500k humanoids shipped in 2030, global fleets in the millions	\$100M+ incremental

\$18.6B
Physical-AI venture funding in Q2 2026 alone (PitchBook)

\$47B
Physical-AI funding in H1 2026 (Crunchbase)

150k+ h
Scale AI physical-AI data engine volume, 10 new robotics customers in 2025

\$4.9B → \$17.1B
Data annotation and training-data market 2025 → 2030 (SCSP)

Sources: company announcements; PitchBook; Crunchbase; Scale AI; SCSP / ISF Voices 2026; SAG (China humanoid forecast). Market construction is AXIOMALITY’s estimate.

COMPETITION

Nobody sells a signed, per-record certificate for robot data; the ingredients are commoditizing, the referee is not

Complements: collection, curation, simulation, runtime safety. Credible threats: NVIDIA adding a manifest; labs staying in-house.

Capability	Collection	Curation / tools	Simulation / world models	Runtime safety	Certification bodies	AXIOMALITY
Formal rulebook, machine-checked consistent	○	○	○	○	○	●
Per-record physics + logic + uncertainty checks	○	◐	◐	◐	○	●
Signed certificate, verifiable offline by anyone	○	○	○	○	◐	●
Cross-embodiment schema (semantic reuse)	○	◐	◐	○	○	●
Audit pack mapped to EU Art. 10/12 and China GB/Z 218.1	○	○	○	◐	●	●
Same rulebook compiled to a runtime guardrail	○	○	○	●	○	●
Who	Scale AI, Mecka, Build AI, XDOF, Human Archive	Foxglove, Encord, Voxel51, Rerun, NVIDIA Cosmos Evaluator (free scoring), Applied Intuition Dana	NVIDIA Cosmos, World Labs Atlas, Lightwheel, Genesis	3Laws, RoboGuard, NVIDIA Halos OS	NVIDIA Halos Inspection Lab (ANAB, TÜV/UL), Pebblous (KR), CR data-set certification (CN)	Kernel + Engine + Passport

● present ◐ partial ○ absent

vs. NVIDIA

Cosmos Evaluator scores plausibility for free but is unsigned, non-formal and tied to the NVIDIA stack; Halos certifies compute and safety. We are the neutral signer at the task-semantics level, on any stack.

vs. in-house pipelines

PI, Figure and DeepMind curate internally. The passport wins where there is information asymmetry: vendors selling to labs, OEMs selling into regulated markets, buyers wanting an independent re-check. Labs are the channel.

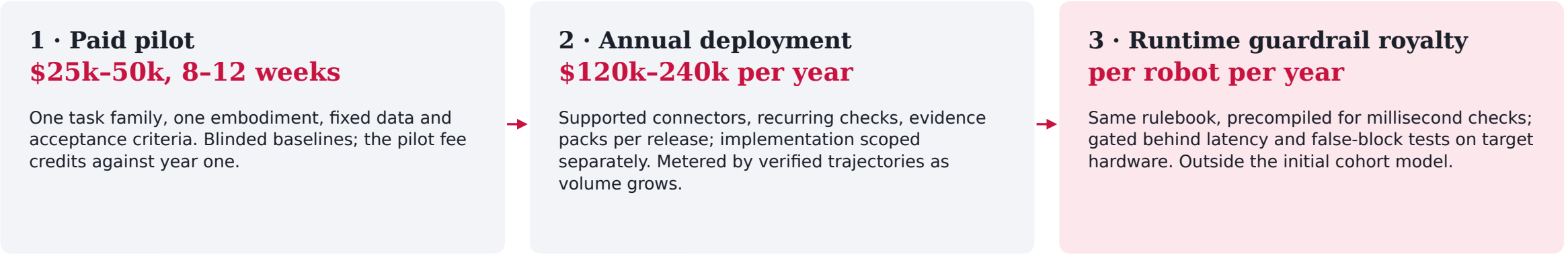
vs. tooling

Foxglove, Rerun and LeRobot are where the data lives; we ship as their “verified” badge, not a replacement. The spec is public; the proven domain rulebook and the signer’s track record do not travel.

Sources: company announcements and press (2025–26); NVIDIA (Mar and Jun 2026); DreamVu 2026 landscape. Positioning is AXIOMALITY’s assessment.

Pilot, then annual deployment, then a per-robot royalty from the same rulebook

Separate bookings, commitments, credits and cash; cohort math below is the sizing example, not a forecast



Sizing cohort (midpoint)	USD	Interpretation
10 pilots × \$40,000	\$400,000	Project bookings
4 annual × \$180,000	\$720,000	Before first-year credits
4 credits × \$20,000	−\$80,000	Avoid double counting
Net software commitment	\$640,000	First contracted year
Combined bookings	\$1,040,000	Pilots + net software

Customer ROI and expansion

Illustrative baseline: 600 review hours a month × \$150 = \$1.08M of annual review capacity; a 40% reduction releases \$432k against a \$180k first-year purchase, before the customer’s own integration and compute. Renewal is the proof, not the model.

Expansion path inside one account: one dataset → every release → a second embodiment → runtime. Gross margin target 80%+ once verification runs in the customer’s environment and compute is theirs.

Source: AXIOMALITY pricing and cohort model; all figures are sizing examples, not contracted.

First buyers own a recurring data decision; the pilot is designed to be falsifiable

Founder-led U.S. and Canadian sales; we pre-register the comparison and keep negative results

Buyer	Budget owner	Initial deliverable	Public workflow examples
Data supplier	Data operations	Repeatable customer acceptance; a passport that raises sale price or shortens buyer review	Scale, Mecka, Human Archive, Build AI
Robot OEM	Autonomy / quality lead	Failure-linked regression evidence; audit pack for EU shipments	Agility / GXO, Sanctuary / Magna, Vention
Model developer	Robot-learning lead	Batch acceptance and diagnostics on mixed real / synthetic data	Physical Intelligence, Figure, Generalist, Dyna
Industrial integrator	Engineering / quality	Application-specific test evidence for a named cell	Universal Robots ecosystem

Pilot design

Same batch through four baselines: the customer’s scripts, simple rules, VLM + rules, full Engine. Expert ground truth built independently; natural and injected errors reported separately.

Acceptance: critical false-accepts, good-data false-rejects, unknown rate, review hours per 1,000 episodes, end-to-end time.

Customer keeps a blind held-out set; model, compute and data budget are fixed.

If the pilot shows...	We...
Simple rules match the full Engine at equal cost	Narrow to the error classes rules miss
Every account needs heavy redesign	Shrink the task domain, price as a service
Augmentation adds no real-world lift	Recalibrate the generator and the mixture
Runtime tail latency fails acceptance	Keep the module offline
Engine beats every baseline and the customer renews	Scale team, scenes and channels

The one number we run on

Verified trajectory hours certified per quarter. Every revenue line sits downstream of it; it measures supply (throughput), demand (who pays for a passport) and quality (what passes) at once.

Sources: AXIOMALITY pilot plan (Sep 2026); public workflow examples from company announcements.

What exists today is assets and relationships; what we will prove in 90 days is three things

Every target carries a date; at least one named design partner moves to “achieved” before the round closes

Achieved

- Knowledge kernel: axioms machine-checked consistent, aligned with ISO/IEC 21838-4 (TUpper / COLORE lineage)
- 100,000 measured 3D packages; room-level reconstruction and object-level 3D annotation stack (9 invention applications)
- Consumer spatial app live on Apple Vision Pro; 13 granted patents
- Beijing International Big Data Exchange data-broker license (China channel)
- Working relationships: an international law firm (clause mapping), a Big Four firm (audit-pack review), multinational medical-device and pharma companies

In progress

- Minimum Engine, internal test Q4 2026: ingest → scene graph → SMT check → Passport
- Kernel v1 for home scenes; connectors for LeRobot v3, RLDS, OpenUSD
- Co-development and channel talks with humanoid OEMs and platform / OS companies

Next 90 days (committed)

- Public blind benchmark: 200 DROID + 200 LeRobot episodes with 7 injected error classes; Engine vs. rules + VLM baseline; per-class recall and cost per hour; a third party holds the injection key
- One named design partner (data supplier or OEM) running real passports on their own data under NDA
- Public Passport spec, public key and CLI verifier with a passport for one open dataset: tamper it, and the verifier fails

Dated targets	Q1 2027	Q2 2027	Month 18	2028
Milestone	3 co-development partners; 10,000 verified episodes	Passport v1 recognized by one assessment body or standards organization	5 cumulative paying customers; 2 renewals or expansions	Audit packs delivered before EU Annex I applies

Why us: we co-wrote the standard, we proved it consistent, and we have run a \$50M-revenue company

Ontology and standards, AI algorithms, perception and robotics, simulation and 3D assets

Founder & CEO

- Standards: core contributor to ISO/IEC 21838-4 (Tupper top-level ontology); member of the Industrial Ontologies Foundry; University of Toronto AI doctorate and researcher
- Science: formal ontology and knowledge representation, consistency proofs and model construction for first-order axiom systems, trustworthy neuro-symbolic AI; 21 papers, FOIS best-paper award, invited AAAI talk, one English-language monograph
- Operating: founding team of a cross-border DTC platform backed by Accel and others (\$5M angel), >\$50M revenue in year one, 100+ person team; co-founded a recommendation-systems company incubated at Toronto and Imperial
- Spatial computing: founded a spatial-computing company, one of the first AR apps on Apple Vision Pro; 13 granted patents, ¥24M patent transfer, ¥10M technology license; prior VC diligence experience

CTO & Chief Scientist

McMaster University AI doctorate; former AI algorithm scientist at a top-tier tech company and startup chief scientist; first place, Canadian AI academic and industrial applications competition

Perception & Robotics Lead

Harbin Institute of Technology doctorate in mechatronics; former AI algorithm engineer and director of the perception center at XPeng; 10+ papers

Simulation & 3D Assets Lead

10+ years as art director in games and stage production; led the digital (VR) reconstruction of 1750 Beijing

Hiring with this round

A robot-learning scientist who has trained VLA policies at scale (owns the benchmark and training-lift experiments); a U.S.-based founding GTM lead for data suppliers and OEMs.

The evidence format for robot data will decide how it is measured, priced and traded.

We are raising a seed round to turn a verified rulebook into the reference implementation: public spec, offline-verifiable passports, no customer data ownership, no foundation model of our own.

Raising

\$4-6M seed

Sized to the 18-month plan with a collection buffer (amount is this revision's suggestion; founders to confirm). Use: core product 45%, independent validation and benchmarks 25%, U.S. GTM and design partners 20%, operations 10%.

1

Months 0-3

Vertical slice: one task, one connector, public benchmark, reproducible evidence package

2

Months 4-9

Two paid validation engagements; all implementation and review effort recorded

3

Months 10-18

Five cumulative customers, two renewals or expansions; second embodiment reused without kernel edits

Three ways to help now: lead the round; introduce one data supplier and one OEM; bring your own data and get a real passport under NDA in a 12-minute session.